

rdf_gsa

BM3R1 启动子强度本身主要决定“延迟/阈值余量”，但系统能不能稳定 toggle，更多取决于 Int 输入残留、Int degradation tag、RDF 表达强度，以及真实 C31 波形质量。

1. BM3R1 延迟指标很干净

bm3r1_delay_h、bm3r1_threshold_margin 基本只由：krep_tsl、K_rep 决定

square 和 c31_waveform 下结果一样

所以：选 BM3R1 promoter strength 时，第一看 krep_tsl，第二看 K_rep

2. 真实功能指标不是单参数控制

在 c31_waveform + bm3r1_promoter 下：toggle_score、switching_contrast、bm3r1_design_score、rdf_mean 都有很高的 ST，但 S1 普遍不高。这说明系统行为主要来自参数组合/交互，不是某一个参数单独决定。

BM3R1 强度需要和 RDF 强度、Int tag、repression threshold 一起匹配

3. square 和 c31_waveform 的区别很重要

square 下 functional outputs 里：k_tag_int、k_int、pulse_width 影响很大，说明理想方波输入时，翻转质量主要受输入强度、输入宽度、Int 残留控制。

c31_waveform 下：k_tag_int、krdf_tsl、krep_tsl、K_rep 更重要，说明真实波形已经固定了输入形状，系统敏感性转移到了 RDF/BM3R1 调控模块上。

4. rdf_waveform 模式最有设计意义

rdf_waveform + c31_waveform 里：baseline_sub、waveform_scale、k_tag_int 对 toggle_score、bm3r1_design_score、int_trough、rdf_mean 都很重要

这说明真实设计里，除了 BM3R1 promoter，还必须控制：C31 baseline leak、C31 waveform amplitude、Int degradation tag

尤其 baseline_sub 和 k_tag_int 对 int_trough 很强，说明降低 Int 残留是关键。

PDF/代码里的 OUTPUT	解释	准确说法
bm3r1_delay_h	BM3R1 关掉 RDF 的延迟时间	由 $\ln(\text{BM3R1_ss} / K_rep) / k_dil$ 估算出来的 repression delay。

PDF/代码里的 OUTPUT	解释	准确说法
bm3r1_design_score	启动子设计综合分数	我们代码里定义的启发式指标，偏好合适 delay、threshold margin，并乘上 toggle 表现。不是原始文献指标。
bm3r1_production	BM3R1 生产速率	$k_{rep_tsl} * mRNA_{ss}$ 。
bm3r1_ss	BM3R1 稳态浓度	$bm3r1_production / k_{dil}$ 。
bm3r1_threshold_margin	BM3R1 阈值余量	$\log(BM3R1_{ss} / K_{rep})$ 。
toggle_score	是否一周期一翻转	采样 LR fraction 后计算的 clean alternation score。
switching_contrast	高低态分离程度	$H_{mean} - L_{mean}$ ，看 LR 高低状态分得清不清楚。
lr_amplitude	LR 动态摆幅	post-transient LR fraction 的 peak-to-peak。
lr_final	最终 LR fraction	仿真终点时 LR DNA fraction。
rdf_max	RDF 总量峰值	total RDF concentration 的最大值。
rdf_mean	RDF 平均残留/平均水平	post-transient total RDF concentration 平均值。
int_peak	Int 峰值	total integrase concentration 的最大值。
int_trough	Int 低谷残留	post-transient total integrase concentration 的最小值。
rep_max	BM3R1 最大浓度	repressor protein 最大值。

"bm3r1_promoter"+"square"

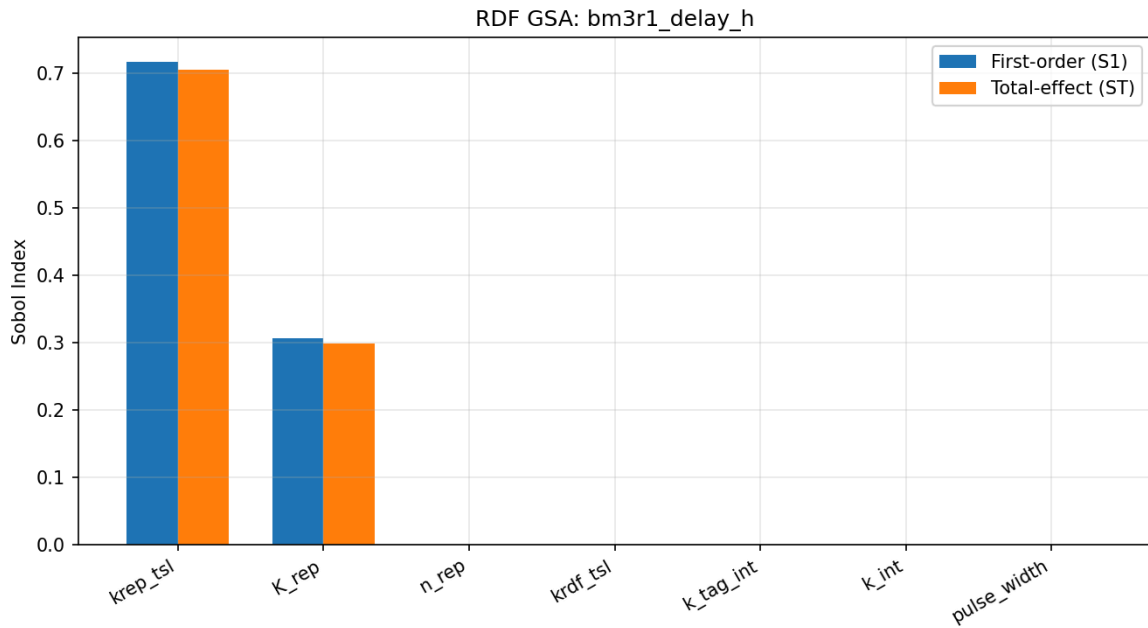
MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "bm3r1_delay_h" 哪些参数决定 BM3R1 关掉 RDF 的延迟时间

===== Sobol Results =====

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krep_tsl    S1 = 0.7175   ST = 0.7053
K_rep       S1 = 0.3067   ST = 0.2980
n_rep       S1 = 0.0000   ST = 0.0000
krdf_tsl    S1 = 0.0000   ST = 0.0000
k_tag_int   S1 = 0.0000   ST = 0.0000
k_int       S1 = 0.0000   ST = 0.0000
pulse_width S1 = 0.0000   ST = 0.0000

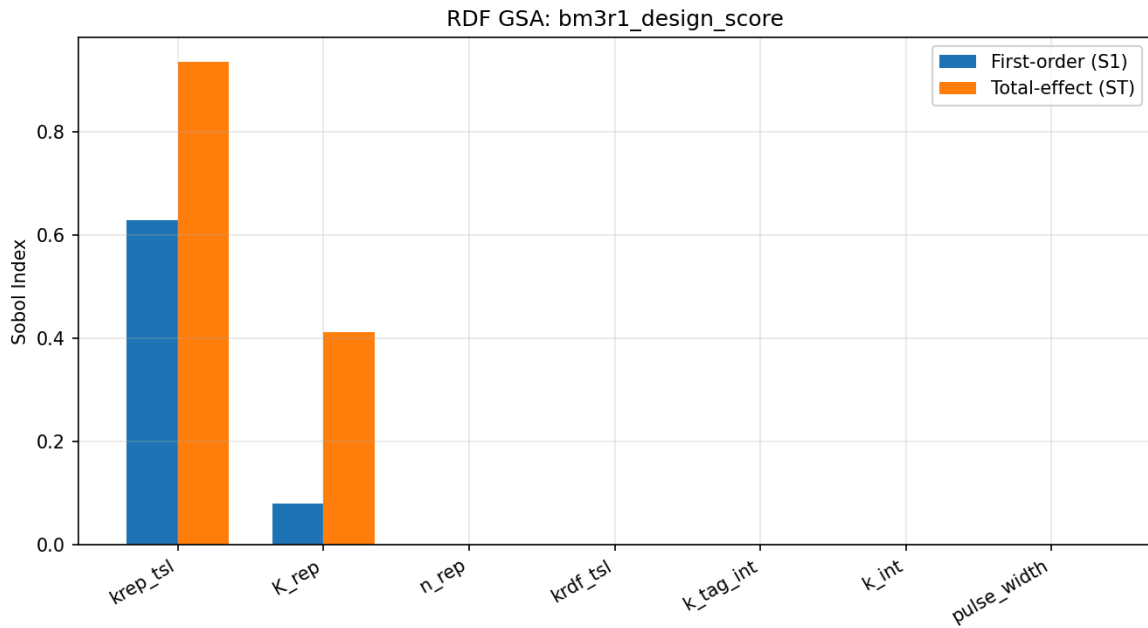
```



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "bm3r1_design_score" 综合指标，偏好合适延迟、足够 threshold margin，以及较好的 toggle 表现。这个更接近“选哪个启动子强度合理”

===== Sobol Results =====

krep_tsl	S1 = 0.6293	ST = 0.9372
K_rep	S1 = 0.0791	ST = 0.4127
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000

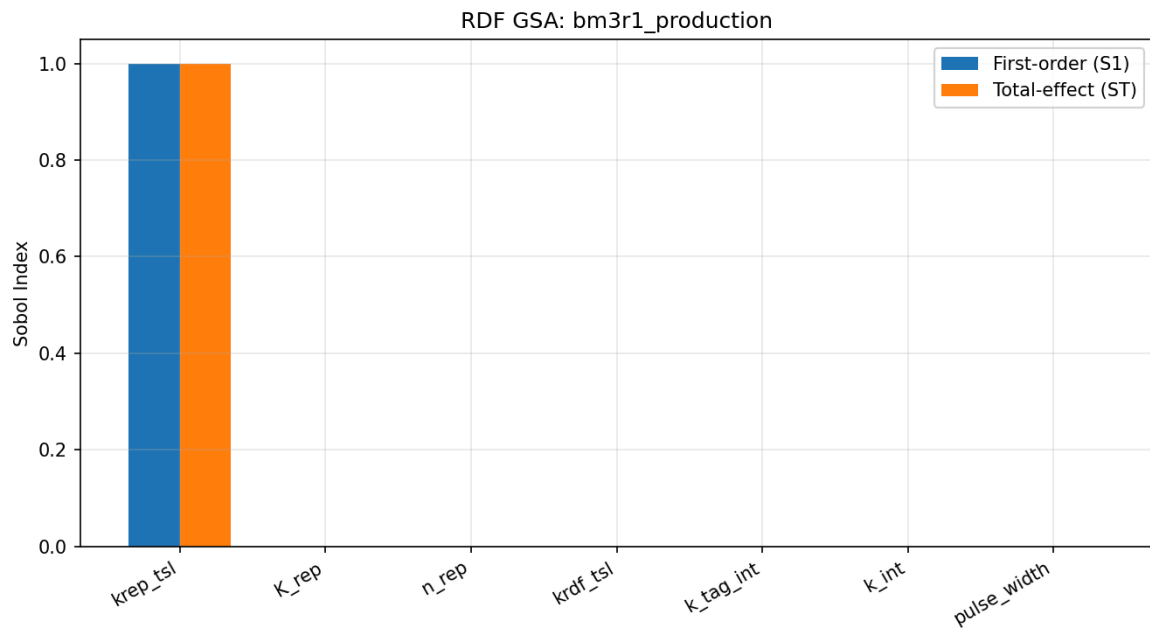


MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "bm3r1_production"

BM3R1 生产速率, 近似对应启动子/RBS 表达强度本身

===== Sobol Results =====

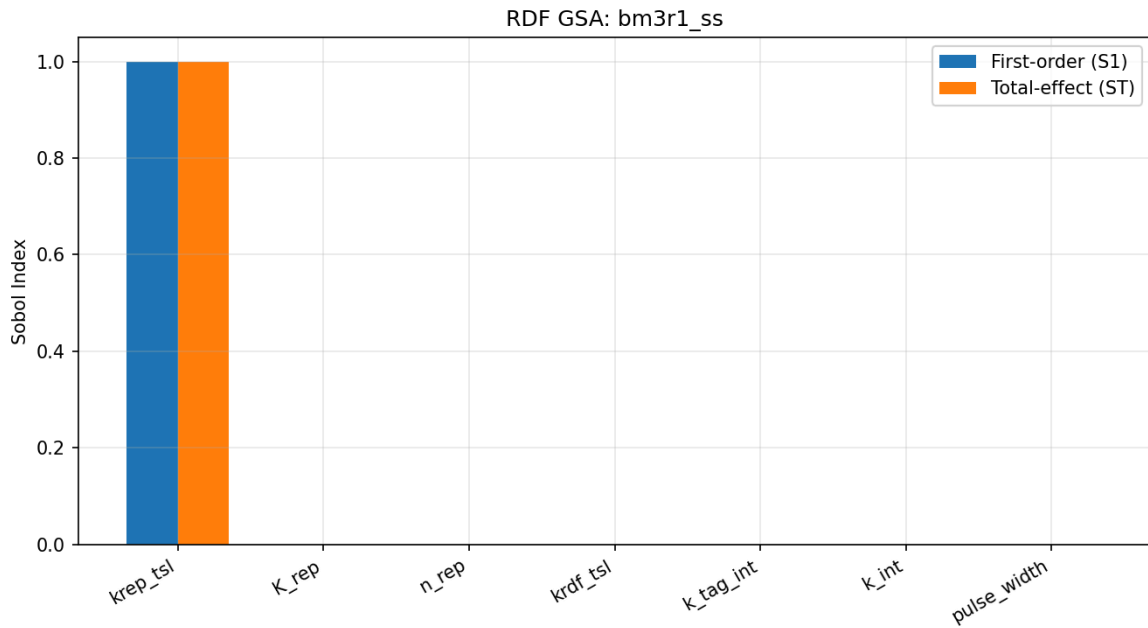
krep_tsl	S1 = 1.0001	ST = 1.0001
K_rep	S1 = 0.0000	ST = 0.0000
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "bm3r1_ss" BM3R1 稳态蛋白浓度

===== Sobol Results =====

krep_tsl	S1 = 1.0001	ST = 1.0001
K_rep	S1 = 0.0000	ST = 0.0000
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000

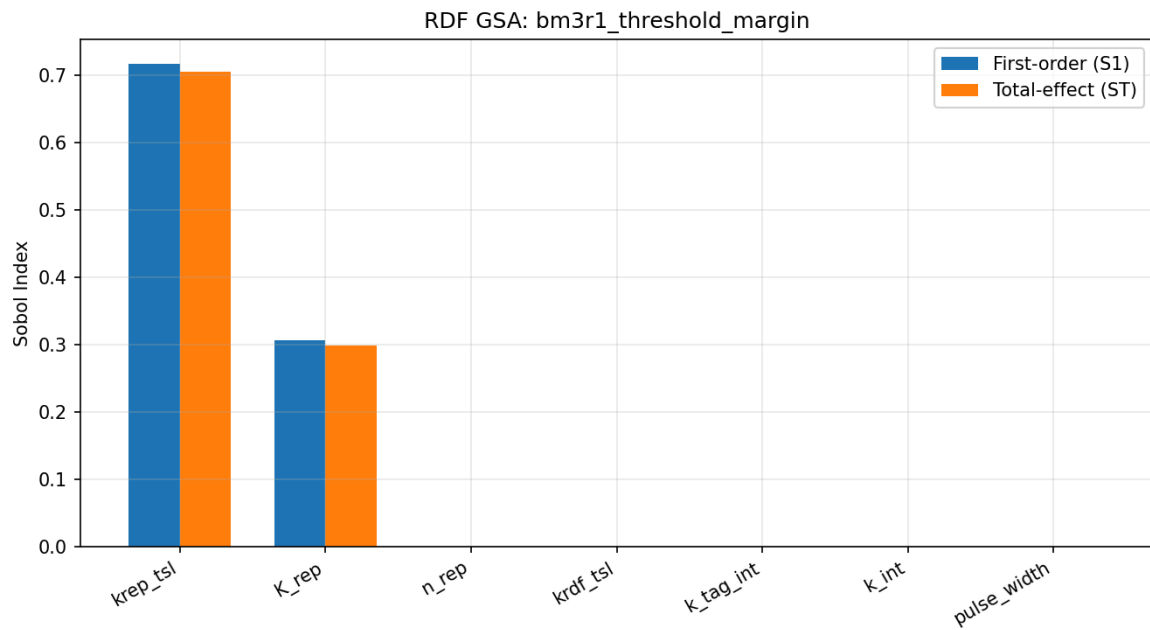


MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT =

"bm3r1_threshold_margin" BM3R1 稳态表达量相对 repression threshold 有多少余量。余量太小会关不住 RDF，太大可能延迟太久

===== Sobol Results =====

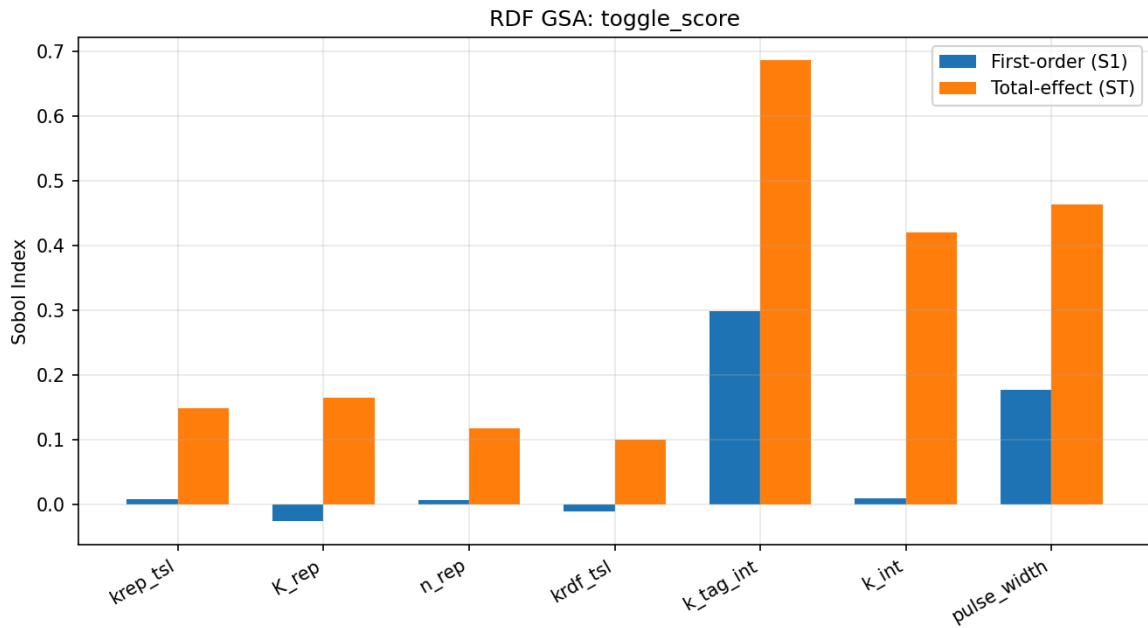
krep_tsl	S1 = 0.7175	ST = 0.7053
K_rep	S1 = 0.3067	ST = 0.2980
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "toggle_score" 系统是否一周期一翻转

===== Sobol Results =====

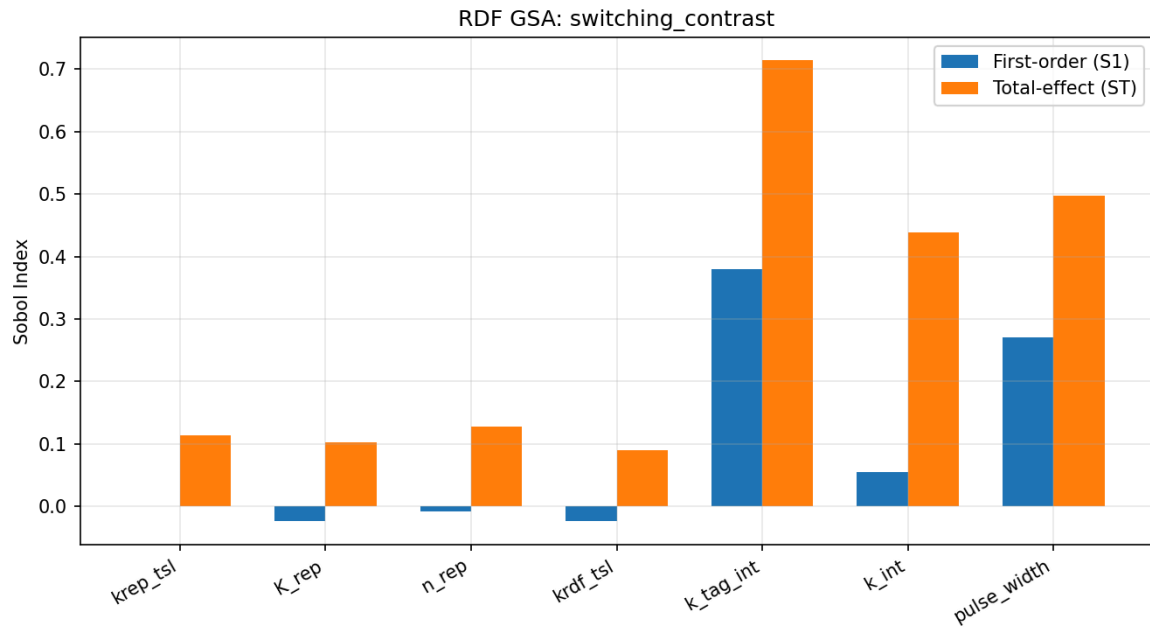
krep_tsl	S1 = 0.0089	ST = 0.1490
K_rep	S1 = -0.0260	ST = 0.1657
n_rep	S1 = 0.0070	ST = 0.1178
krdf_tsl	S1 = -0.0105	ST = 0.1006
k_tag_int	S1 = 0.2986	ST = 0.6866
k_int	S1 = 0.0096	ST = 0.4209



MODE = "bm3r1_promoter" SOURCE_MODE = "square" switching_contrast 看 LR 高低态分离是否清楚。高低态差距越大，计数器状态越可靠

===== Sobol Results =====

krep_tsl	S1 = 0.0008	ST = 0.1137
K_rep	S1 = -0.0236	ST = 0.1032
n_rep	S1 = -0.0081	ST = 0.1275
krdf_tsl	S1 = -0.0228	ST = 0.0905
k_tag_int	S1 = 0.3794	ST = 0.7137
k_int	S1 = 0.0552	ST = 0.4387
pulse_width	S1 = 0.2704	ST = 0.4973



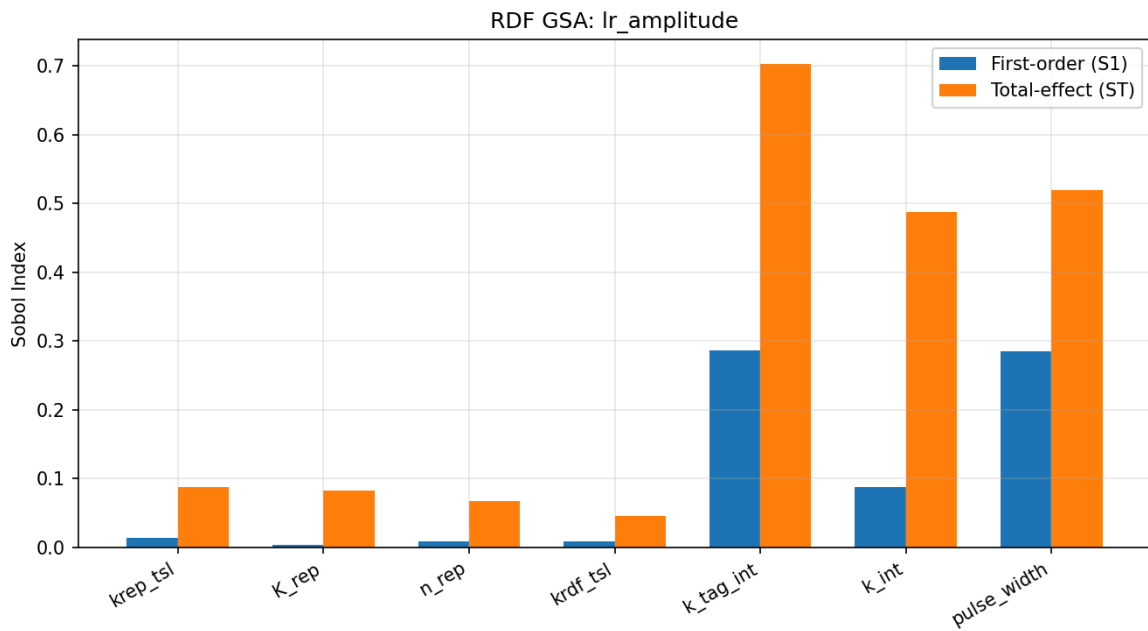
MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "lr_amplitude" LR

fraction 的动态摆幅, 辅助判断有没有明显翻转

===== Sobol Results =====

krep_tsl	S1 = 0.0138	ST = 0.0870
K_rep	S1 = 0.0030	ST = 0.0823
n_rep	S1 = 0.0084	ST = 0.0669
krdf_tsl	S1 = 0.0087	ST = 0.0459
k_tag_int	S1 = 0.2868	ST = 0.7034
k_int	S1 = 0.0876	ST = 0.4881

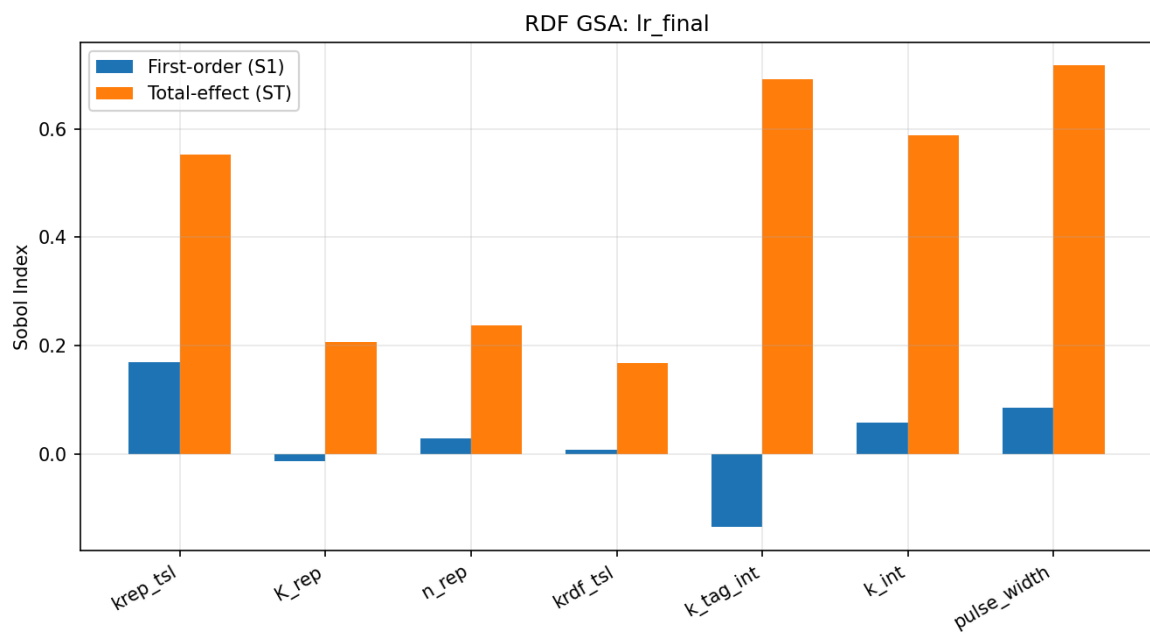
pulse_width S1 = 0.2849 ST = 0.5200



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "lr_final" 看最终系统停在哪个 DNA 状态，但它不等价于“翻转好不好”，只能作辅助

===== Sobol Results =====

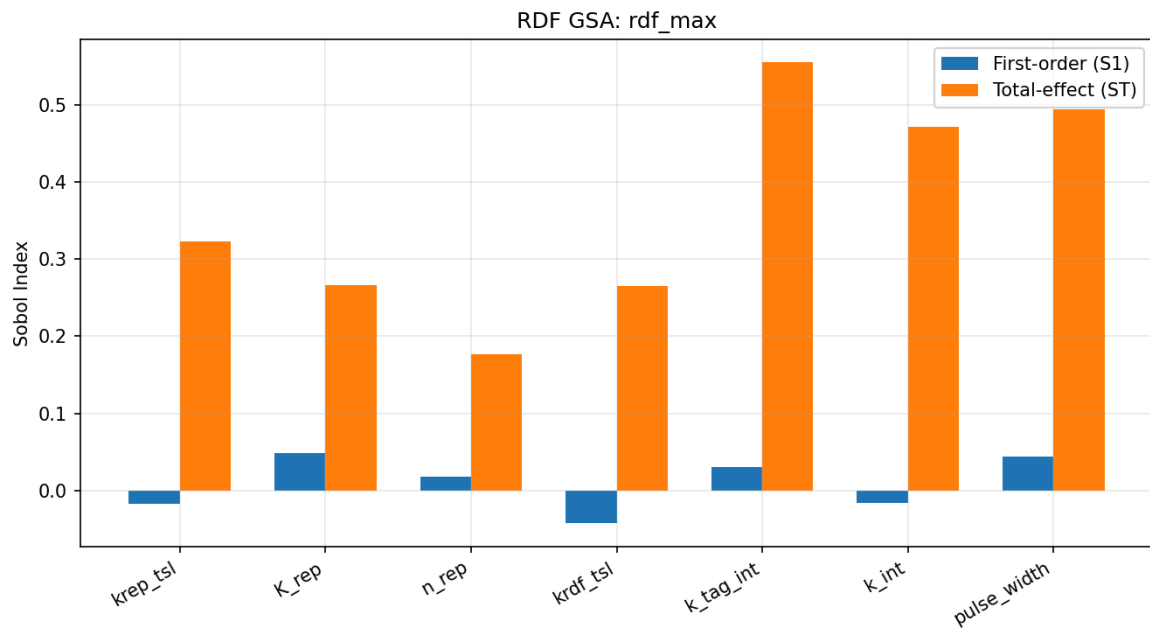
krep_tsl	S1 = 0.1699	ST = 0.5530
K_rep	S1 = -0.0131	ST = 0.2058
n_rep	S1 = 0.0288	ST = 0.2373
krdf_tsl	S1 = 0.0078	ST = 0.1673
k_tag_int	S1 = -0.1352	ST = 0.6912
K_int	S1 = 0.0576	ST = 0.5883
pulse_width	S1 = 0.0844	ST = 0.7167



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "rdf_max" RDF 总量峰值

===== Sobol Results =====

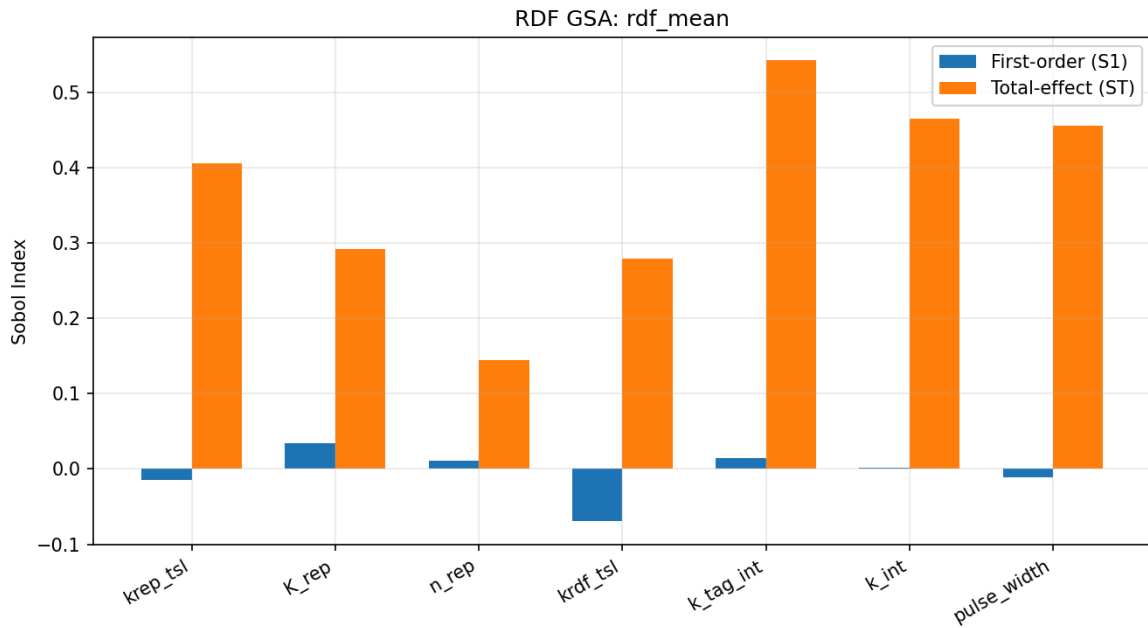
krep_tsl	S1 = -0.0173	ST = 0.3223
K_rep	S1 = 0.0493	ST = 0.2659
n_rep	S1 = 0.0187	ST = 0.1766
krdf_tsl	S1 = -0.0422	ST = 0.2654
k_tag_int	S1 = 0.0307	ST = 0.5546
k_int	S1 = -0.0156	ST = 0.4709
pulse_width	S1 = 0.0447	ST = 0.4938



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "rdf_mean"

===== Sobol Results =====

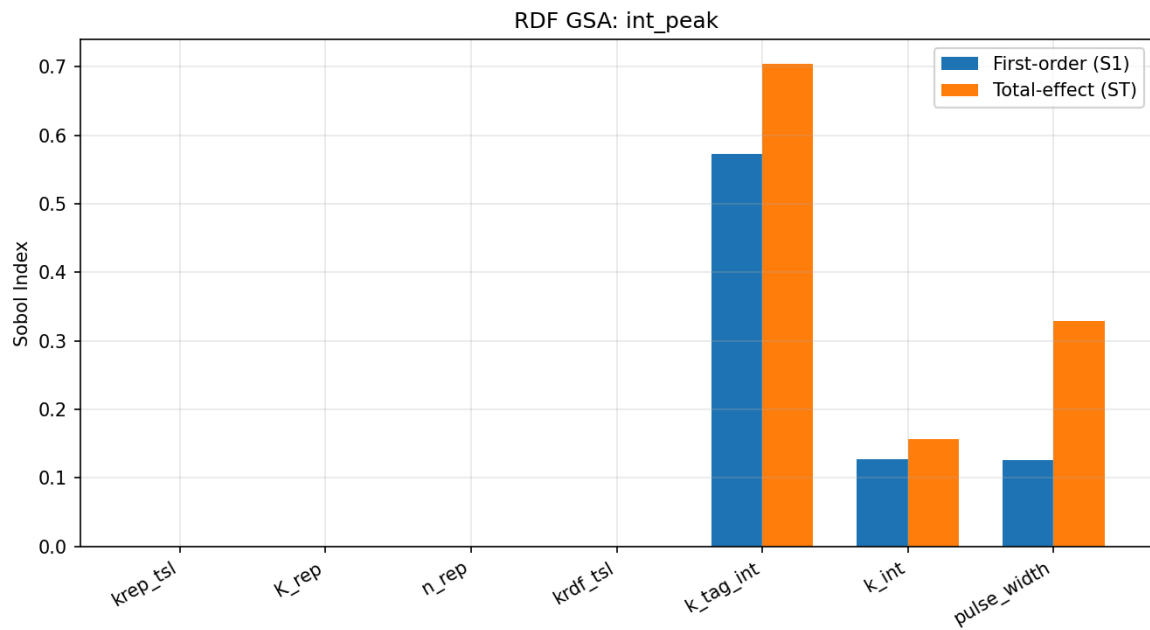
krep_tsl	S1 = -0.0143	ST = 0.4058
K_rep	S1 = 0.0347	ST = 0.2917
n_rep	S1 = 0.0110	ST = 0.1441
krdf_tsl	S1 = -0.0696	ST = 0.2787
k_tag_int	S1 = 0.0149	ST = 0.5424
k_int	S1 = 0.0017	ST = 0.4649
pulse_width	S1 = -0.0116	ST = 0.4560



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "int_peak" integrase 输入
峰值，决定翻转驱动力是否足够

===== Sobol Results =====

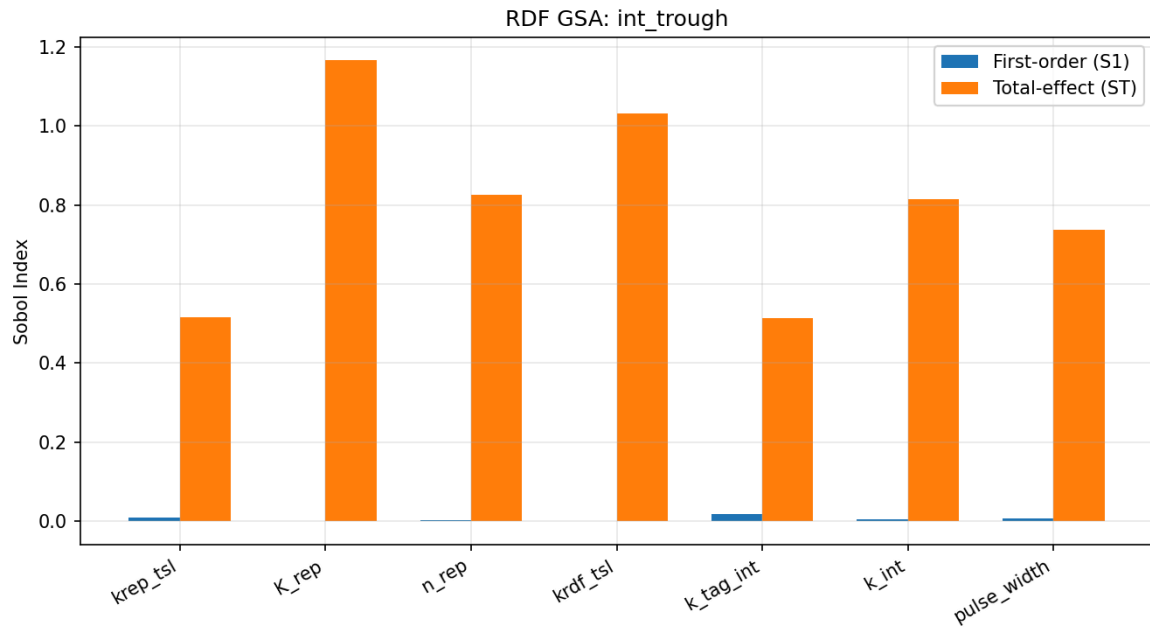
krep_tsl	S1 = 0.0000	ST = 0.0000
K_rep	S1 = 0.0000	ST = 0.0000
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = -0.0000	ST = 0.0000
k_tag_int	S1 = 0.5731	ST = 0.7049
k_int	S1 = 0.1270	ST = 0.1568
pulse_width	S1 = 0.1258	ST = 0.3292



MODE = "bm3r1_promoter" SOURCE_MODE = "square" OUTPUT = "int_trough" integrase 低
谷残留

===== Sobol Results =====

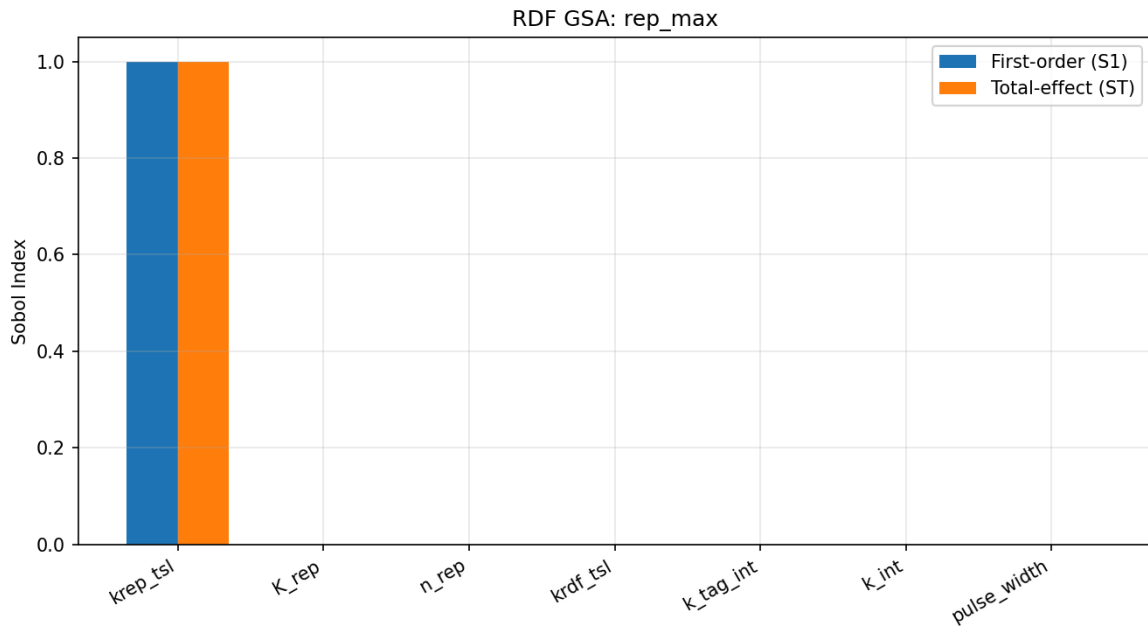
krep_tsl	S1 = 0.0094	ST = 0.5158
K_rep	S1 = -0.0000	ST = 1.1657
n_rep	S1 = 0.0040	ST = 0.8255
krdf_tsl	S1 = 0.0001	ST = 1.0310
k_tag_int	S1 = 0.0177	ST = 0.5138
k_int	S1 = 0.0054	ST = 0.8139
pulse_width	S1 = 0.0072	ST = 0.7371



MODE = "bm3r1_promoter" SOURCE_MODE = "square" BM3R1 最大浓度

===== Sobol Results =====

krep_tsl	S1 = 1.0001	ST = 1.0001
K_rep	S1 = 0.0000	ST = 0.0000
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000

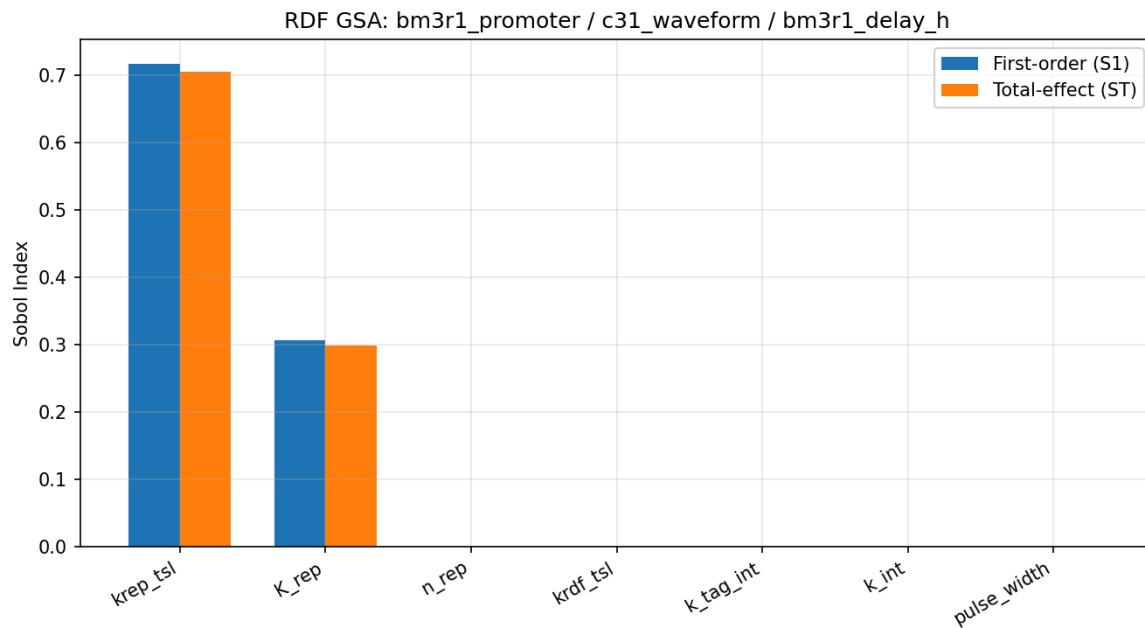


"bm3r1_promoter"+"c31_waveform"

MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" "bm3r1_delay_h"

===== Sobol Results =====

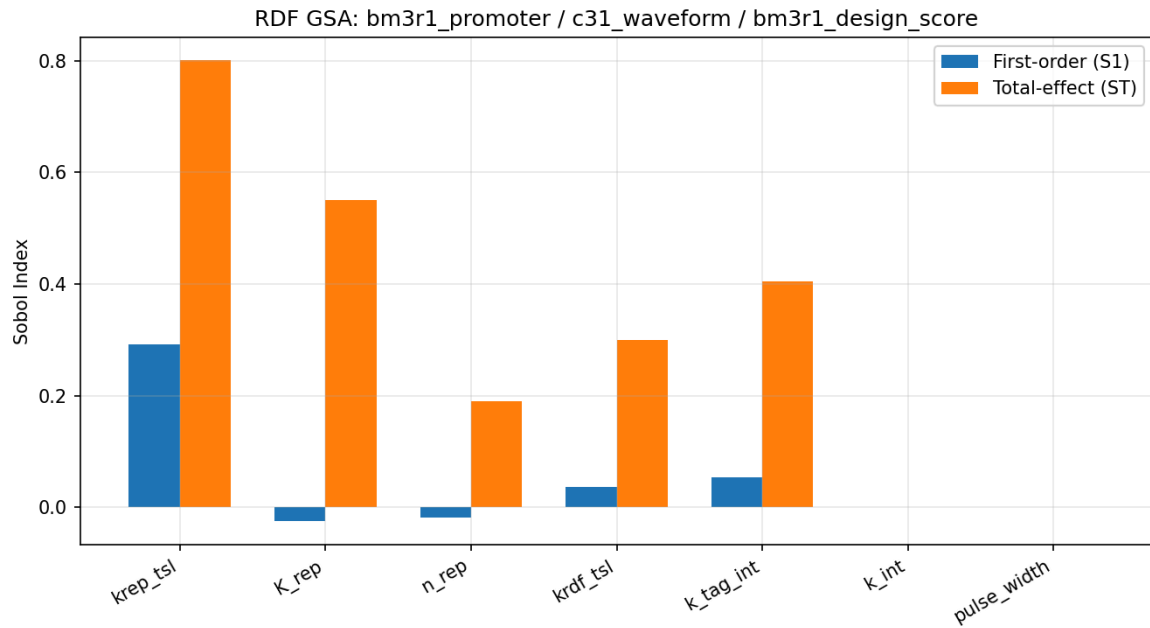
krep_tsl	S1 = 0.7175	ST = 0.7053
K_rep	S1 = 0.3067	ST = 0.2980
n_rep	S1 = 0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 0.0000	ST = 0.0000
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT =
 "bm3r1_design_score"

===== Sobol Results =====

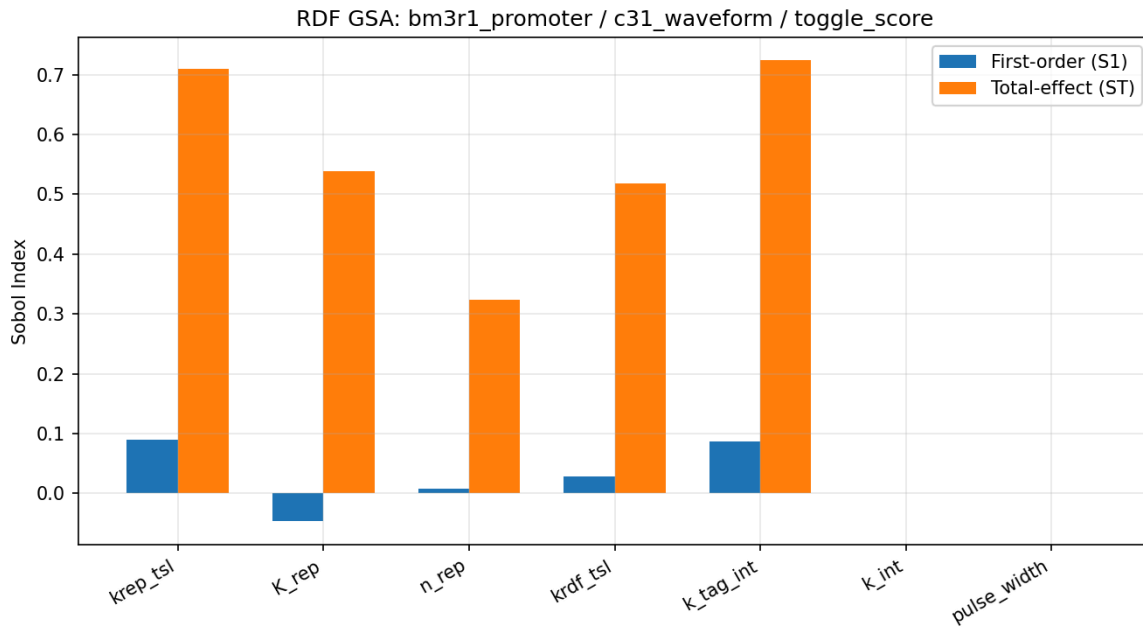
krep_tsl	S1 = 0.2913	ST = 0.8003
K_rep	S1 = -0.0248	ST = 0.5503
n_rep	S1 = -0.0178	ST = 0.1899
krdf_tsl	S1 = 0.0364	ST = 0.2997
k_tag_int	S1 = 0.0536	ST = 0.4050
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT = "toggle_score"

===== Sobol Results =====

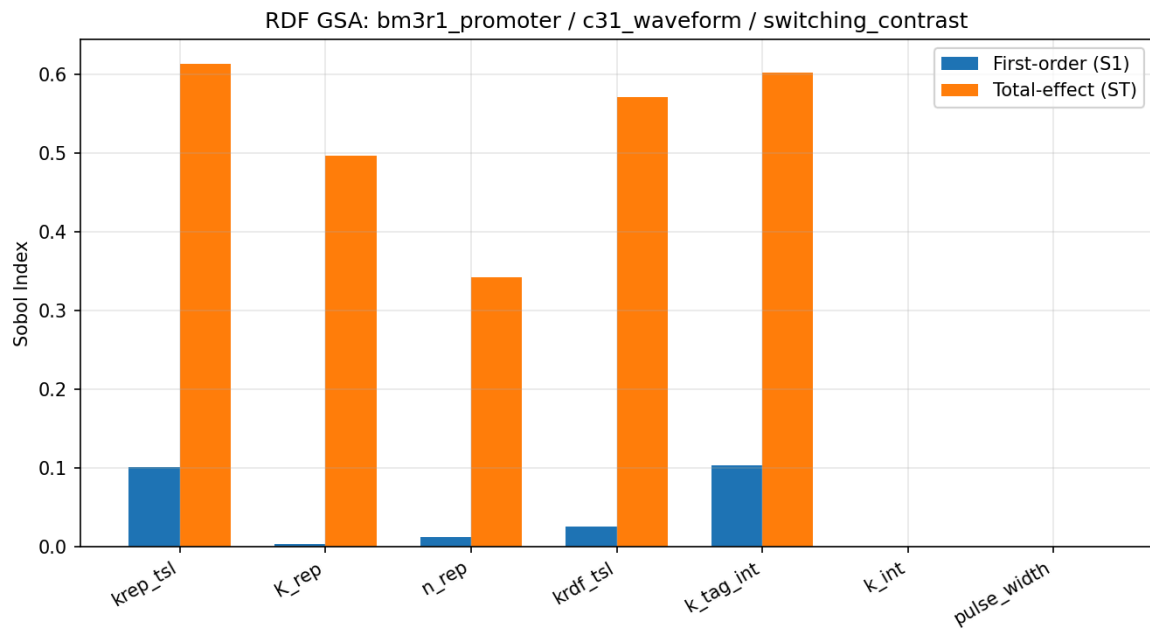
krep_tsl	S1 = 0.0892	ST = 0.7102
K_rep	S1 = -0.0469	ST = 0.5394
n_rep	S1 = 0.0081	ST = 0.3240
krdf_tsl	S1 = 0.0277	ST = 0.5176
k_tag_int	S1 = 0.0861	ST = 0.7240
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT =
"switching_contrast"

===== Sobol Results =====

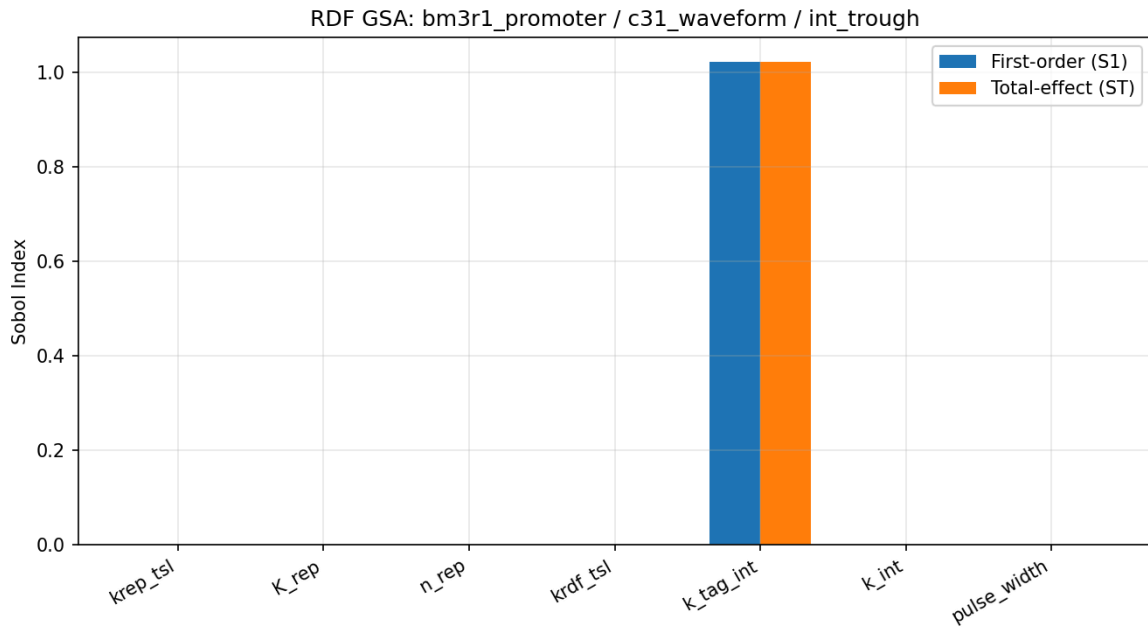
krep_tsl	S1 = 0.1005	ST = 0.6135
K_rep	S1 = 0.0030	ST = 0.4968
n_rep	S1 = 0.0123	ST = 0.3420
krdf_tsl	S1 = 0.0251	ST = 0.5703
k_tag_int	S1 = 0.1031	ST = 0.6018
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT = "int_trough"

===== Sobol Results =====

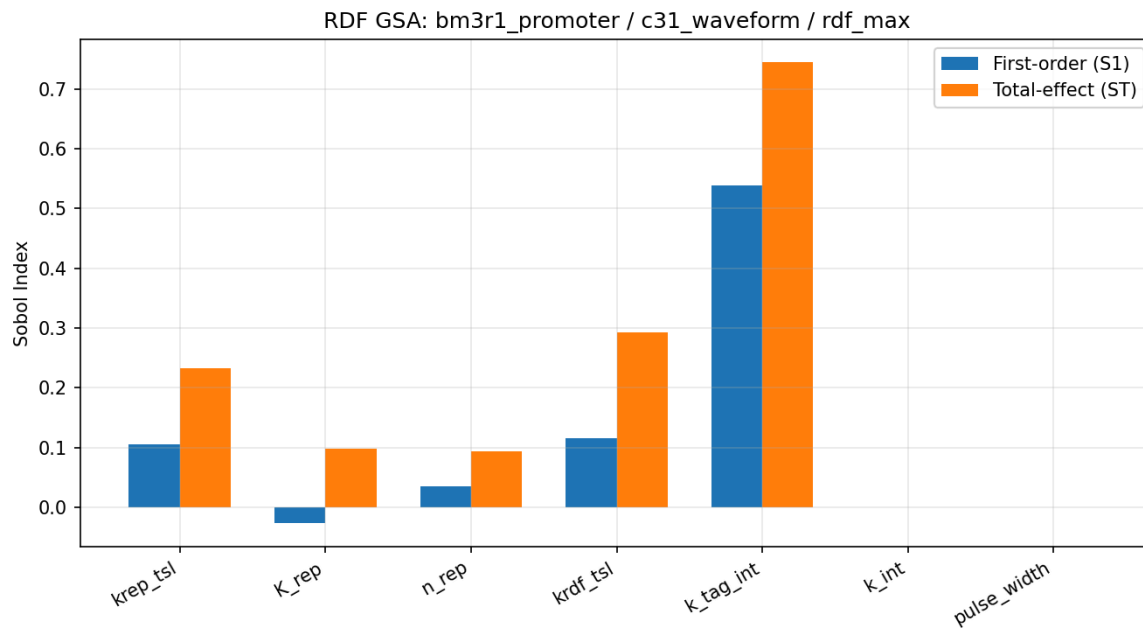
krep_tsl	S1 = 0.0000	ST = 0.0000
K_rep	S1 = -0.0000	ST = 0.0000
n_rep	S1 = -0.0000	ST = 0.0000
krdf_tsl	S1 = 0.0000	ST = 0.0000
k_tag_int	S1 = 1.0233	ST = 1.0233
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT = "rdf_max"

===== Sobol Results =====

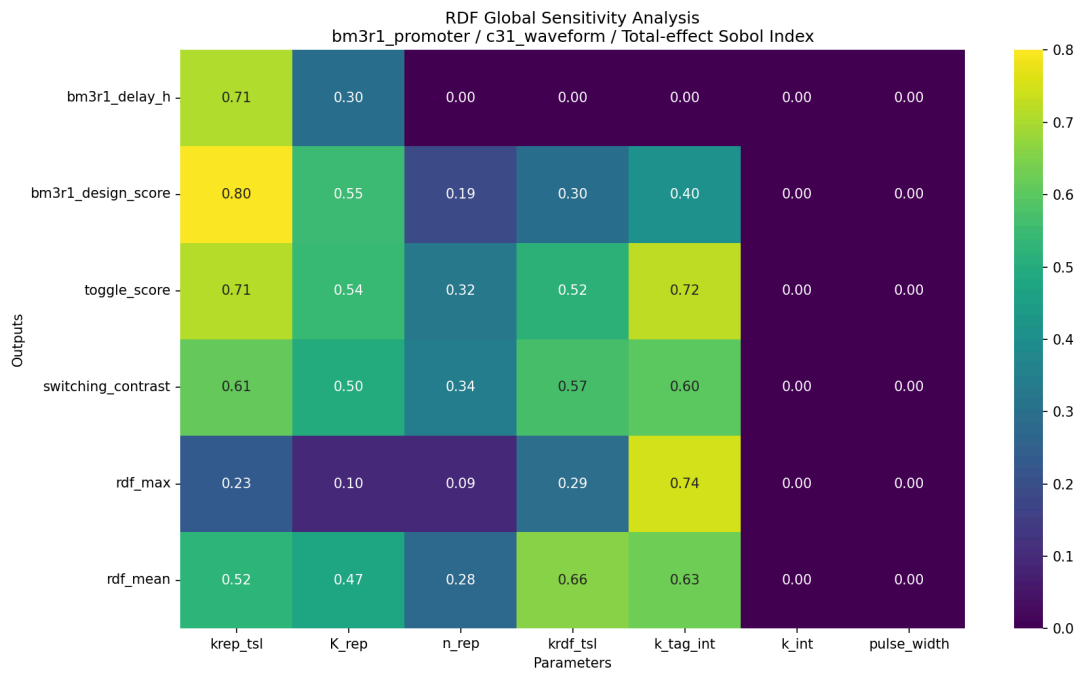
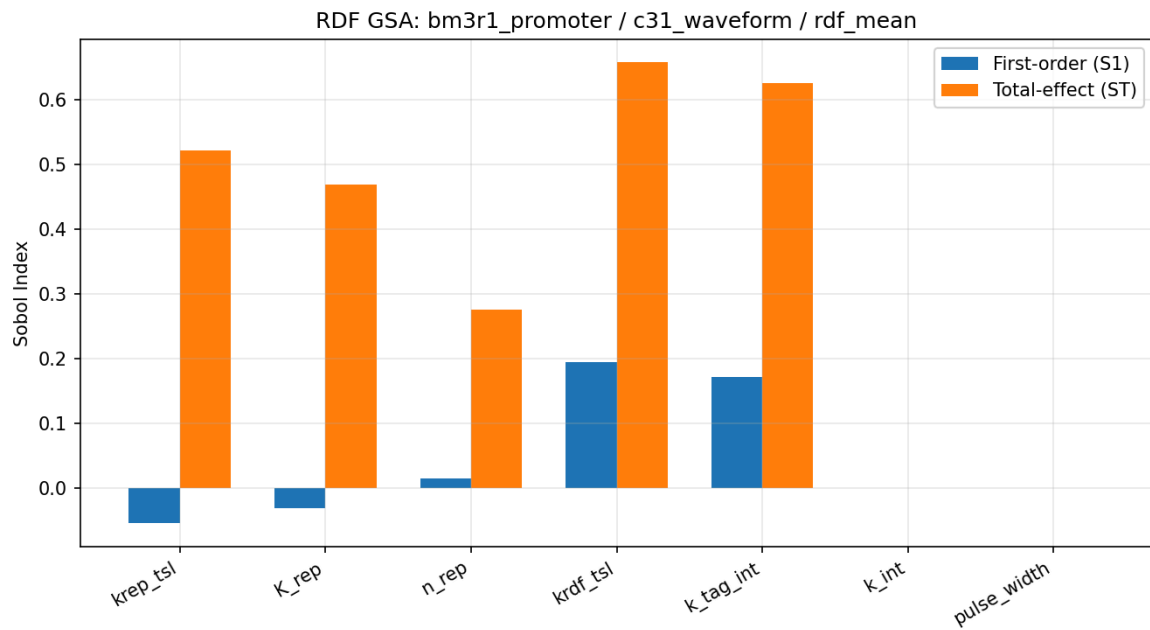
krep_tsl	S1 = 0.1046	ST = 0.2325
K_rep	S1 = -0.0270	ST = 0.0986
n_rep	S1 = 0.0350	ST = 0.0938
krdf_tsl	S1 = 0.1154	ST = 0.2920
k_tag_int	S1 = 0.5386	ST = 0.7447
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000



MODE = "bm3r1_promoter" SOURCE_MODE = "c31_waveform" OUTPUT = "rdf_mean"

===== Sobol Results =====

krep_tsl	S1 = -0.0548	ST = 0.5221
K_rep	S1 = -0.0309	ST = 0.4689
n_rep	S1 = 0.0150	ST = 0.2759
krdf_tsl	S1 = 0.1951	ST = 0.6578
k_tag_int	S1 = 0.1714	ST = 0.6264
k_int	S1 = 0.0000	ST = 0.0000
pulse_width	S1 = 0.0000	ST = 0.0000

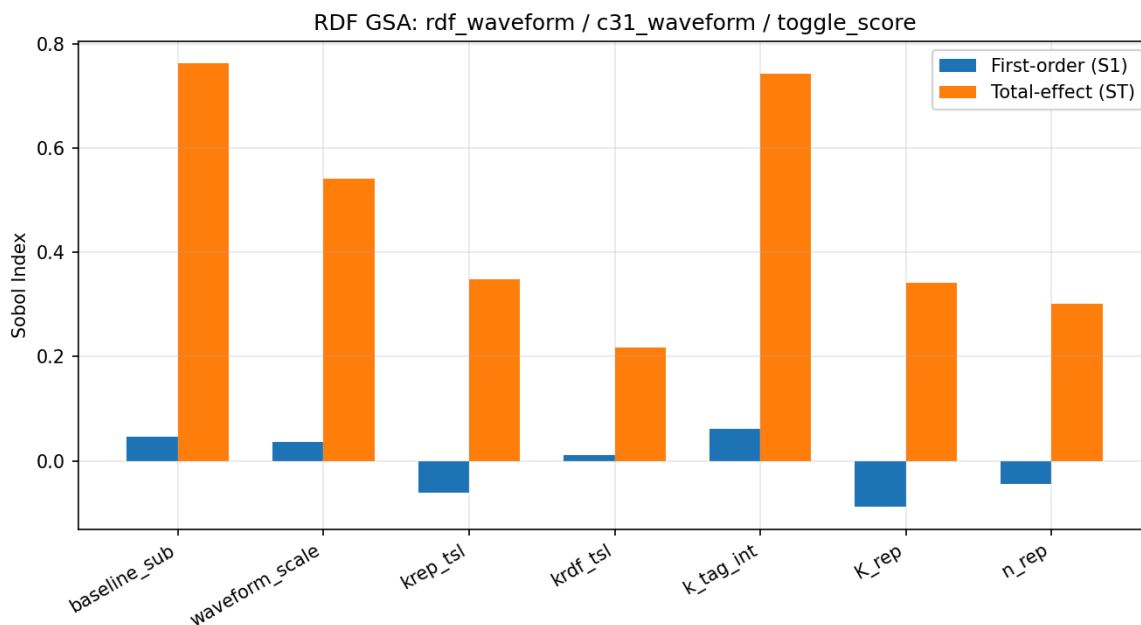


"rdf_waveform"+"c31_waveform"

MODE = "rdf_waveform" SOURCE_MODE = "c31_waveform" OUTPUT = "toggle_score"

===== Sobol Results =====

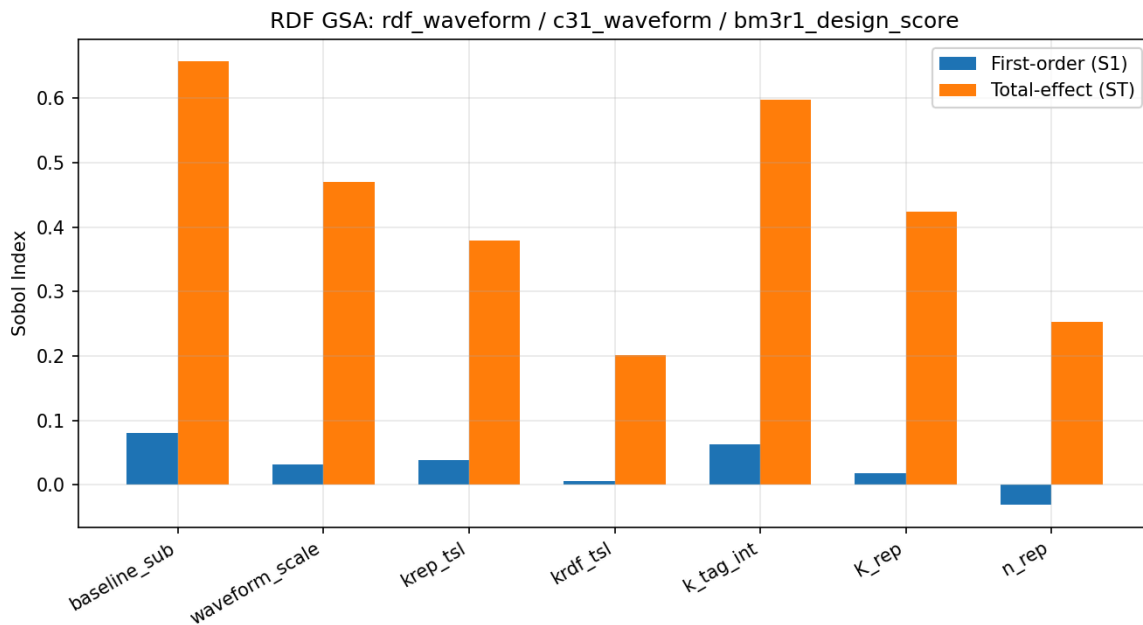
baseline_sub S1 = 0.0465 ST = 0.7629
waveform_scale S1 = 0.0354 ST = 0.5416
krep_tsl S1 = -0.0611 ST = 0.3487
krdf_tsl S1 = 0.0107 ST = 0.2177
k_tag_int S1 = 0.0618 ST = 0.7436
K_rep S1 = -0.0885 ST = 0.3414
n_rep S1 = -0.0453 ST = 0.3020



MODE = "rdf_waveform" SOURCE_MODE = "c31_waveform" OUTPUT = "bm3r1_design_score"

===== Sobol Results =====

baseline_sub S1 = 0.0805 ST = 0.6569
waveform_scale S1 = 0.0320 ST = 0.4697
krep_tsl S1 = 0.0381 ST = 0.3785
krdf_tsl S1 = 0.0062 ST = 0.2015
k_tag_int S1 = 0.0633 ST = 0.5974
K_rep S1 = 0.0175 ST = 0.4242
n_rep S1 = -0.0310 ST = 0.2535

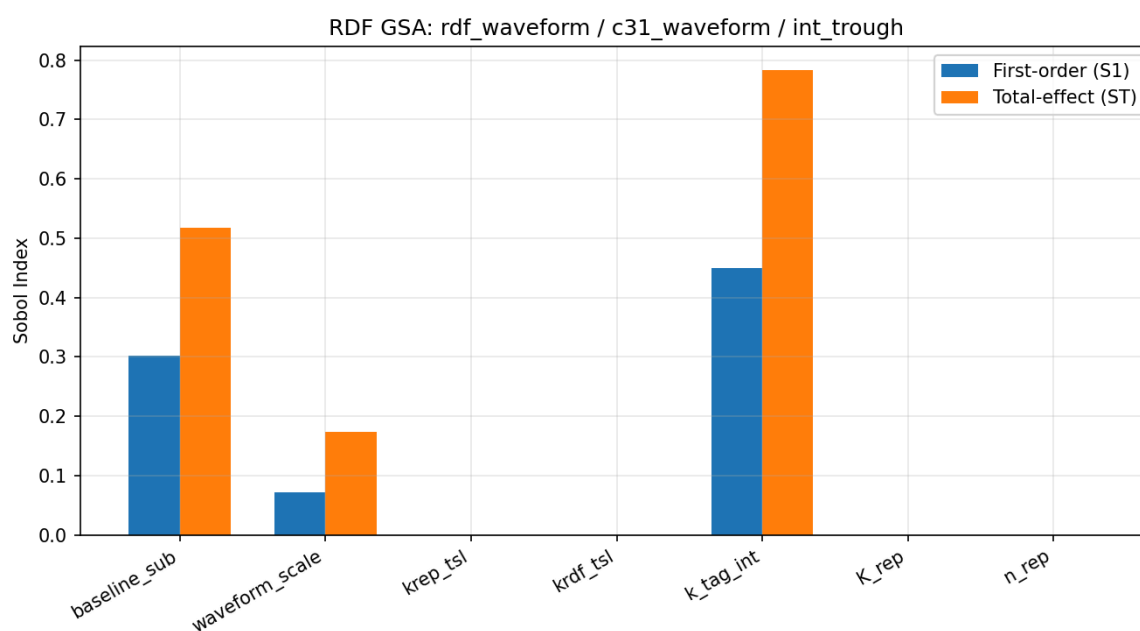


MODE = "rdf_waveform" SOURCE_MODE = "c31_waveform" OUTPUT = "int_trough"

===== Sobol Results =====

baseline_sub S1 = 0.3022 ST = 0.5177
 waveform_scale S1 = 0.0714 ST = 0.1730
 krep_tsl S1 = 0.0000 ST = 0.0000
 krdf_tsl S1 = -0.0000 ST = 0.0000
 k_tag_int S1 = 0.4491 ST = 0.7837
 K_rep S1 = -0.0000 ST = 0.0000

n_rep S1 = -0.0000 ST = 0.0000



MODE = "rdf_waveform" SOURCE_MODE = "c31_waveform" OUTPUT = "rdf_mean"

===== Sobol Results =====

baseline_sub S1 = 0.0113 ST = 0.8925

waveform_scale S1 = 0.0714 ST = 1.3025

krep_tsl S1 = -0.0139 ST = 0.5294

krdf_tsl S1 = 0.0570 ST = 0.6160

k_tag_int S1 = 0.0111 ST = 0.9453

K_rep S1 = -0.0150 ST = 0.4859

n_rep S1 = 0.0443 ST = 0.3513

